RAJALAKSHMI ENGINEERING COLLEGE

K Nithin 251001169-2025-IT K2

Started on	Friday, 7 August 2026, 8:35 AM
State	Finished
Completed on	Friday, 7 August 2026, 9:13 AM
Time taken	37 mins 52 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
{
    int i= 1;
```

```
    int s =1;
```

```
    while(s <= n)
    {
         i++;
         s += i;
     }
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

For example:

Input	Result
9	12

Answer: (penalty regime: 0 %)

```
#include <stdio.h>
int main()
{
    int n;
    scanf("%d", &n);
    int i = 1;
    int s = 1;
    int counter = 2;
    while (1){
        counter++;
        if (s <= n){
            i++;
            counter++;
            s += i;
            counter++;
        }
        else{
            break;
        }
    }
    printf("%d", counter);
    return 0;
}
```

	Input	Expected	Got	
✔	9	12	12	✔
✔	4	9	9	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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